

Addis Ababa University
Faculty of Informatics
Department of Computer Science

Entrance Examination for the M. Sc. Program in Computer Science

Date: March 14, 2005
Time allowed: 3 hours

Name: _____

Instructions

1. Read the instructions and the questions carefully.
2. Write your name only on this page. Writing your name on the other pages will automatically disqualify you from this exam.
3. Answer all questions eligibly on the spaces provided on the question papers. Please make sure that sentences are short, grammatically and logically correct. You can use the back side of each paper when space is not sufficient. In such a case, please indicate so.
4. This examination booklet consists of 10 pages (excluding this cover page). Make sure that the question papers are numbered consecutively and that no question papers are missing or replicated.
5. Cheating is strictly prohibited. Any person found cheating will be automatically disqualified from this exam.
6. Use only the rough papers provided. Using other unauthorized papers will be considered as cheating.

For Office Use Only

	Part	Part I (35)	Part II (25)	Part III (20)	Part IV (20)	Total (100)
	Marks					

Part I: Programming, Data Structures, and Software Engineering (35 points)

1. Discuss the basic components of an SRS (Software Requirements Specification). (6 pts)

2. Explain the difference between a DFD and structure chart. (3 pts)

3. Answer the following questions.

a. Define a binary tree. (2 pts)

b. Explain array representation of a binary tree. Illustrate your answer by an example. (4 pts)

c. What are binary tree traversals? Give their recursive algorithms. (6 pts)

4. For each of the following three multiple choice questions, circle the letter of the correct answer. (1 pt each)
- i. Which of the following is true about structured programming or structured problem solving?
 - a. The problem is divided into a set of time dependent/sequential statements that determine which part is executed how many times and how long.
 - b. The problem is divided into a set of functionally independent units.
 - c. The language gives emphasis to data structures needed in solving the given problem.
 - d. All except c.
 - ii. Which of the following is true about recursive problem solving?
 - a. The problem will have a known solution if each iteration (calling) leads the problem to a case where we know how to solve.
 - b. Even if the solution is inefficient because of performance and storage reasons, it is recommended as it increases the readability of the program than the iterative counter part.
 - c. It is recommended as it is not possible to solve some problems iteratively.
 - d. All of the above.
 - iii. In Ethiopia, every citizen is called with his/her name rather than his/her family's name as is the case in most other countries. It is possible that the name of a person, say, "XYZ" to be the same as the name of his/her grandfather. In a gathering where only children exist, the name XYZ is meant to refer to the child rather than the grandfather. Which of the following scope rule represents the above hypothesis?
 - a. XYZ is a global name, it is used equally everywhere without ambiguity.
 - b. XYZ is a Local name.
 - c. XYZ can be both global and local. When it is used in the local context, it is given priority.
 - d. None of the above.
5. In modular programming, functions communicate through parameters (arguments). Parameter passing by value is recommended to improve the performance of a system than using parameter passing by reference. Why? (3 pts)

6. In enterprise applications development, securing data from disclosure to unauthorized users is one of the challenges. One means of tackling the challenge is to encrypt before sending and decrypting at the receiving end with the same key. Write two functions Encrypt and Decrypt that accept a string and a key so as to encrypt and decrypt the text. The encryption algorithm adds 1 to the ASCII value of each character and subtracts one from the ASCII value of each character for decryption. (8 pts)

Hint: if one wants to send “Hello”, it is “Ifoop” that is sent and when decrypted it will be Hello.

Part II: Operating Systems and Computer Architecture (25 points)

1. Describe three features which might make one process scheduling algorithm preferable to another. (3 pts)
2. What distinguishes a good page replacement algorithm from a bad page replacement algorithm? (3 pts)
3. In general terms, give two different ways of solving the problem of deadlock. (2 pts)
4. When a UNIX process forks a child process, name one thing the parent and child processes have in common, and two things the parent and child processes don't have in common. (3 pts)
5. Give two advantages of RAM over hard disk, and two advantages of hard disk over RAM. (2 pts)

6. What is the basic advantage of using interrupt-initiated data transfer over transfer under program control without an interrupt? (4 pts)
7. The purpose of the communication link is to resolve the differences that exist between the central computer and each peripheral. List any two such differences. (4 pts)
8. Compare and contrast physical segments and logical segments in Memory Address Space (MAS). (4 pts)

Part III: Database Management Systems (20 points)

1. How do you tackle a database designing problem that involves many-to-many relationship in the context of RDBMSs? (4 pts)

2. The Ethiopian Athletics Federation wants to keep track of the records of Ethiopian athletes who are taking part in International competitions. To this effect, the following relations are identified:

Athlet_Info(AID, FName, SName, BirthDate), where AID refers to the ID of an athlete,

Comptetion_Info(CID, AID, Cdescription, PlaceHeld, DateHeld, TimeToFinsih, Distance), where CID refers to an ID of a competition, and

RecordHolder_Info(AID, CID)

Entries in the first two tables are mandatory, whereas entries in the third table may or may not exist for an athlete as it depends on whether an athlete has won a competition with a new record or not.

Based on the above information answer the following questions:

- a. Write a DDL statement to create the table named RecordHolder_Info. (3 pts)
- b. Are all the above relations in 2NF? Justify your answer. (3 pts)

- c. Write an SQL statement that computes the total number of records held by an athlete. (2 pts)

- d. Write an SQL statement that finds the best time of an athlete on a specific distance out of the competitions that he/she has taken part. (3 pts)

- e. Write an SQL statement that returns a query output containing list of names of athletes who took part in the competitions held in “Boston”, USA. (2 pts)

- 3. Explain what new or emerging class(es) of applications that RDBMSs have failed to cope with or manage. (3 pts)

Part IV: Networking (20 points)

- [illegible]

